

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
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Register Number	192524427

MODEL PRACTICAL EXAMINATION – SET 1

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud web application for Student Counselling System using any Cloud Service Provider to demonstrate SaaS. Include the fields of Student Name, Age, Address, Phone, Subjects Studied, Year of Passing, Percentage of Marks, Domain Interest, etc.
2. Create a simple cloud software application for Library Book Reservation System for SIMATS library using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Title of the Book, Author, Publications, Year, Edition, No. of Copies, Rack No., Status (Taken/Returned), etc.
3. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.
4. Showcase virtual machine migration based on certain conditions, from one host to another.

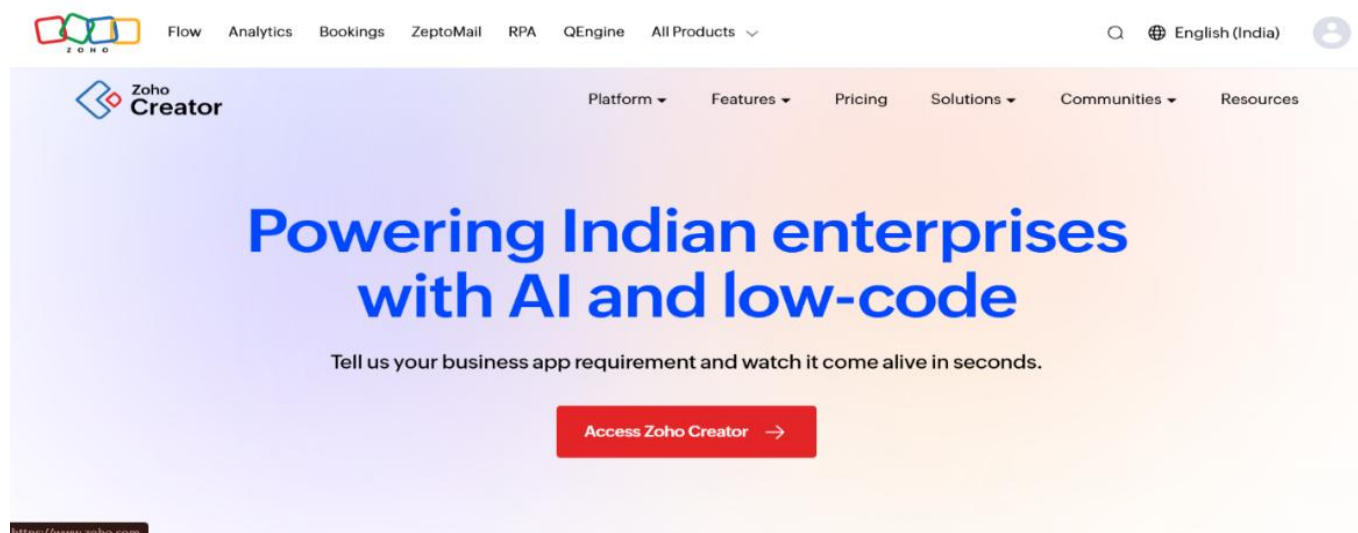
1. Create a simple cloud web application for Student Counselling System using any Cloud Service Provider to demonstrate SaaS. Include the fields of Student Name, Age, Address, Phone, Subjects Studied, Year of Passing, Percentage of Marks, Domain Interest, etc.

AIM:

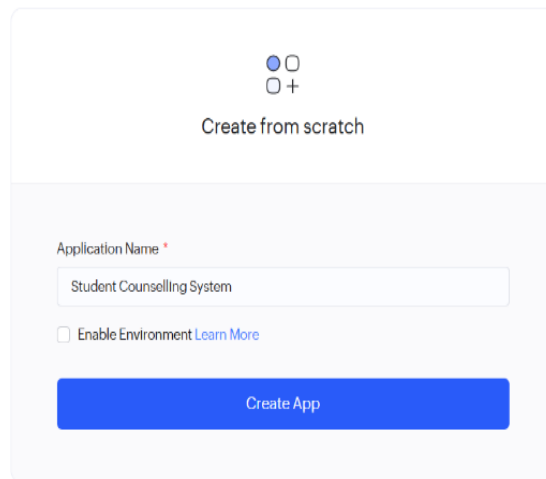
To Create a simple cloud web application for student counselling system using any cloud service provider to demonstrate SAAS with the fields of student name, age, address, phone, subjects studied, year of passing, percentage of marks, domain interest, etc.

PROCEDURE:

STEP 1: Log into **zoho.com**.

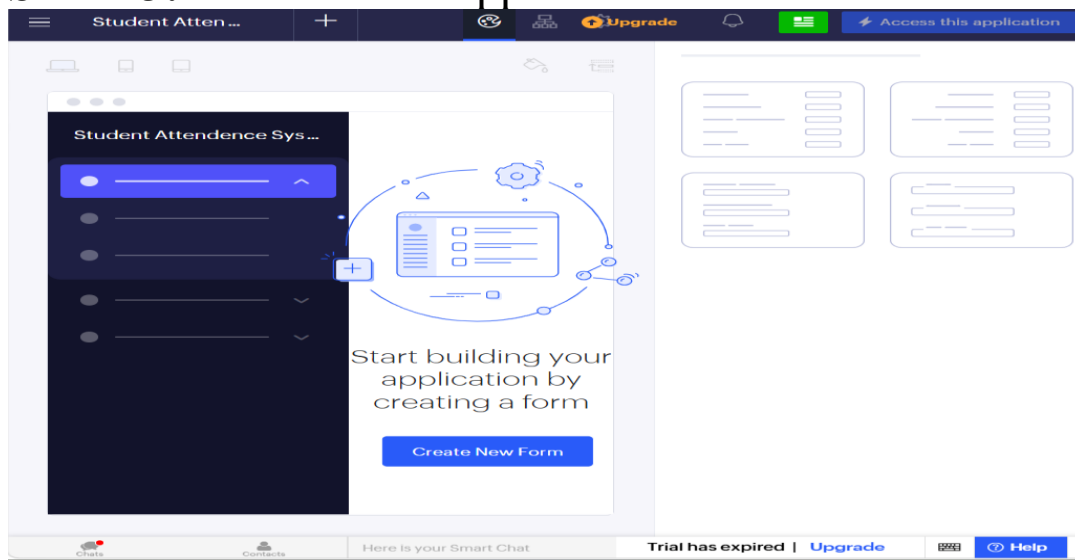


STEP 2: Create a new application from scratch in Zoho Creator.



The screenshot shows the 'Create from scratch' interface in Zoho Creator. At the top, there are two icons: a blue circle with a white square inside, and a white square with a plus sign. Below these icons is the text 'Create from scratch'. The main form area has a label 'Application Name *' followed by a text input field containing 'Student Counselling System'. Below the input field is a checkbox labeled 'Enable Environment' with a link 'Learn More' next to it. At the bottom of the form is a large blue button labeled 'Create App'.

STEP 3: Create a new application and create forms from scratch



STEP 4: Create Student form.

%

Percent

\$

Currency

.00

Decimal

SubForm

Signature

Audio

1

2

Auto Number

Date-Time

Lookup

File upload

Video

f(x)

Formula

Name

Phone

Address

Number

DOB

Email

Drop Down

Percent

Field Properties

Field name

Name

Field link name

Name

View Field References

Validation

Mandatory

Display Fields

Prefix

First Name

Last Name

Suffix

Data Privacy

Here is your Smart Chat (Ctrl+Space)

Trial has expired | Upgrade

Help

STEP 5: Create Academic Form

SIMATS Mark Sheet Processing S...
Students

+

Done

Basic Fields

Name

Email

Address

Phone

Single Line

Multi Line

Number

Date

Student ID

Student Name

Register Number

Department

Year

Section

Mobile Number

Email

Field Properties

Field name

Student ID

Field link name

student_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

STEP 6: Create subject Form

student counselling system

Usage Metrics

fields



Name



Email



Address



Phone



Single Line



Multi Line



Number



Date



Time



Drop Down

Student

Plan

123 Login Count

123 Records Created

123 Reports Generated

123 Storage Used

Last Activity

Metric Date

STEP 7: Create Domain Form

Basic Fields



Name



Email



Address



Phone



Single Line



Multi Line



Number



Date



Time



Drop Down

Student

Primary Domain

Secondary Domain

Interest Level

Career Goal

Recommended Courses

Analysis Date

Analyst Notes

STEP 8: Add sample data to all the created forms.

DashboardSubject HistoriesStudent ProfilesDomain Preferences

Domain Preference

Student

-Select-

Primary Domain

-Select-

Secondary Domain

-Select-

Tertiary Domain

-Select-

Preference Rank

#####

Interest Level

-Select-

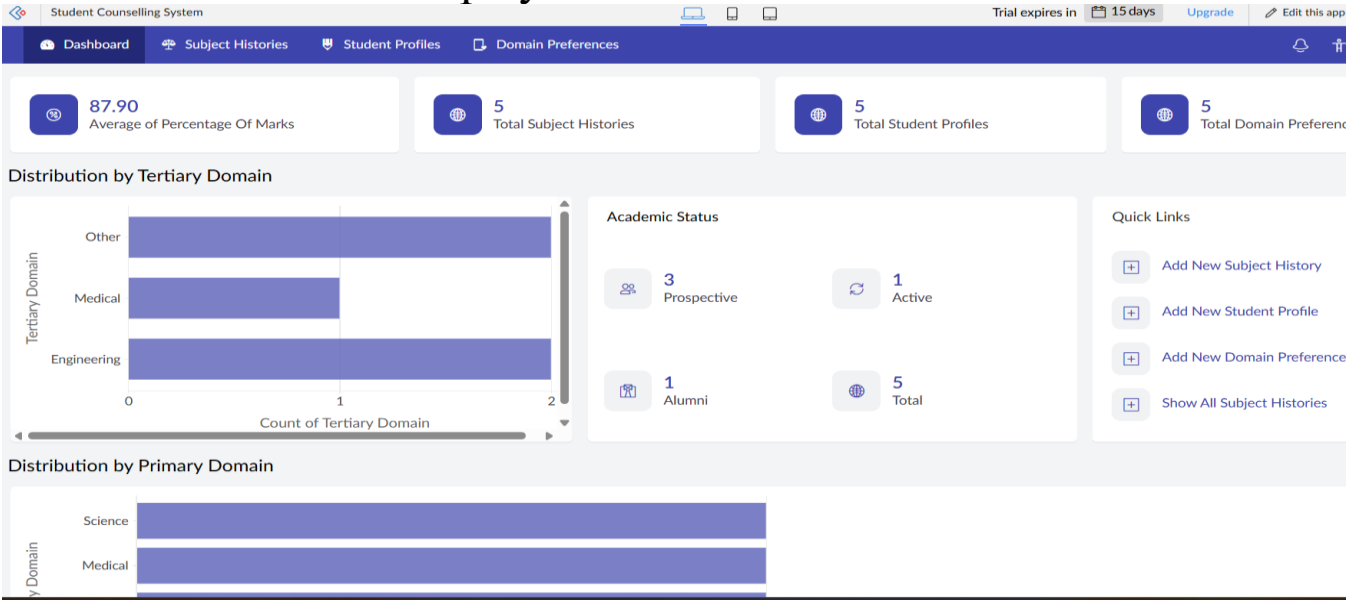
Career Goal

Notes

Submit

Reset

STEP 9: Create and display the Dashboard



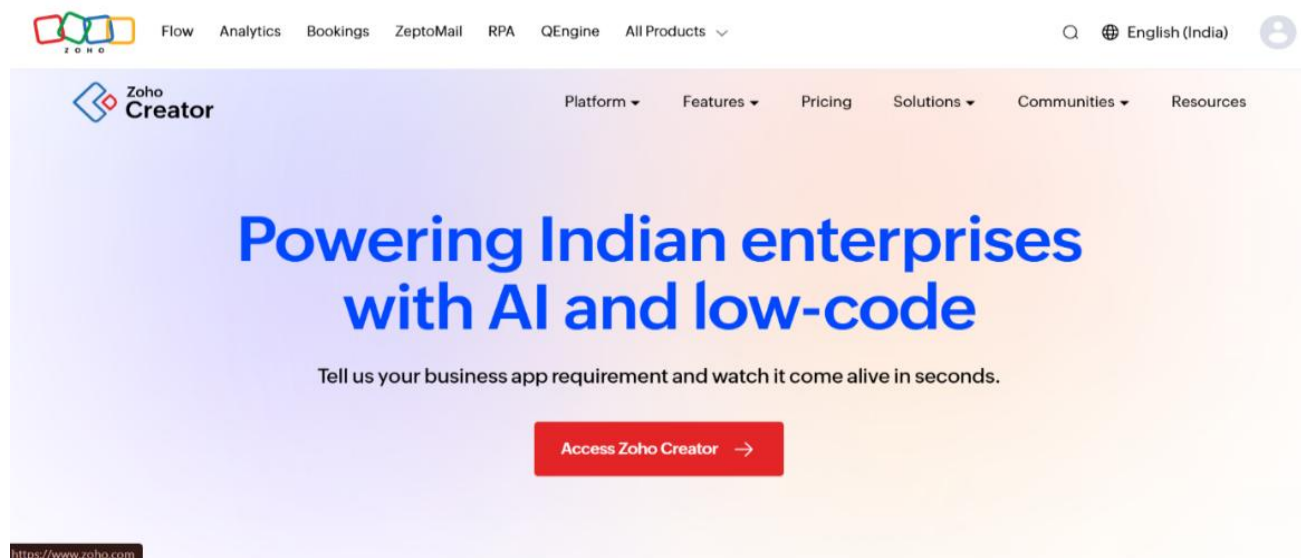
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AIM:

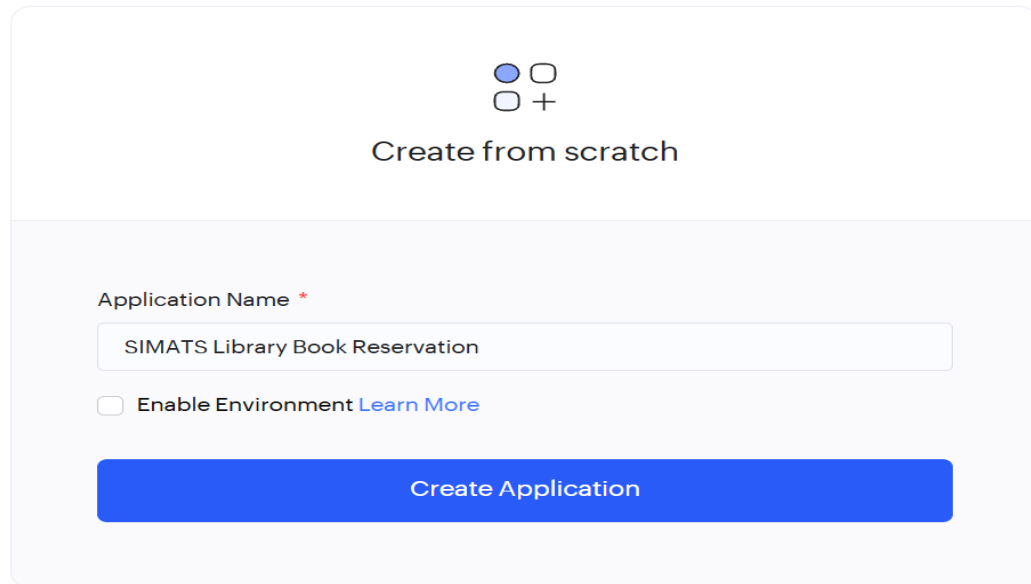
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PROCEDURE:

STEP 1: Log into [zoho.com](https://www.zoho.com).

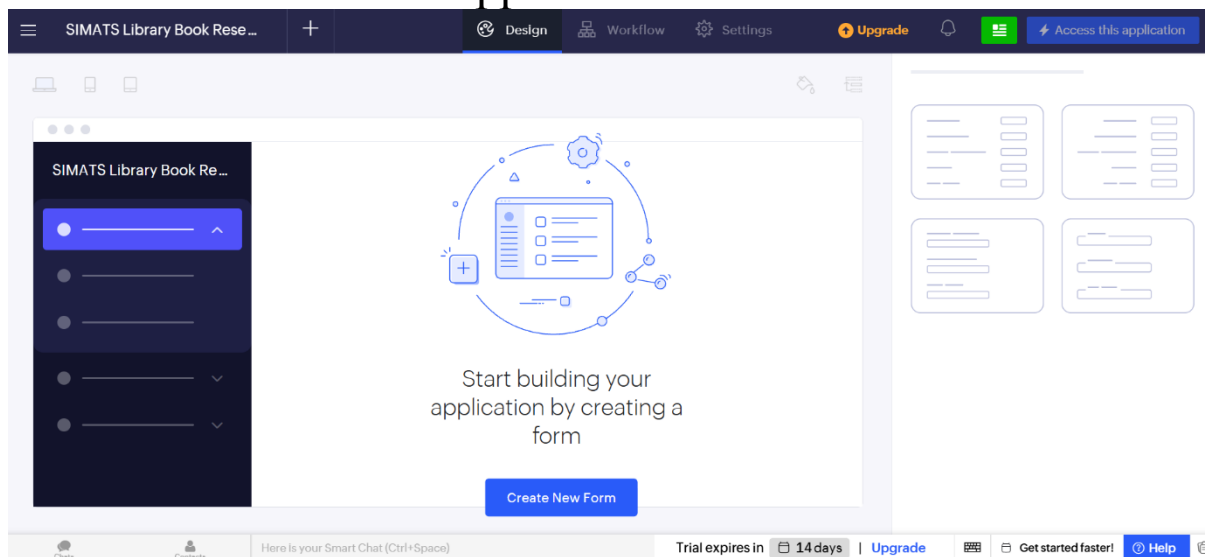


STEP 2: Create a new application from scratch in Zoho Creator.



The screenshot shows the 'Create from scratch' interface in Zoho Creator. At the top, there is a logo consisting of four squares (two blue, two white) arranged in a 2x2 grid, with a plus sign below them. Below the logo, the text 'Create from scratch' is centered. Underneath, there is a section titled 'Application Name' with a red asterisk. Below this title is a text input field containing the text 'SIMATS Library Book Reservation'. Below the input field, there is a checkbox labeled 'Enable Environment' followed by a link 'Learn More'. At the bottom of the section is a large blue button labeled 'Create Application'.

STEP 3: Create a new application and create forms from scratch.



The screenshot shows the Zoho Creator application design interface. The top navigation bar includes a menu icon, the application name 'SIMATS Library Book Re...', a plus icon, and tabs for 'Design', 'Workflow', and 'Settings'. There are also buttons for 'Upgrade', a notification bell, a list icon, and a button labeled 'Access this application'. The main workspace is divided into three sections. On the left is a dark sidebar with the application name and a list of items. In the center is a large white area with a circular diagram showing a workflow and a central box with a plus sign. Below the diagram, the text 'Start building your application by creating a form' is displayed, followed by a blue button labeled 'Create New Form'. On the right is a panel with four form templates. The bottom status bar includes a chat icon, a contacts icon, a message 'Here is your Smart Chat (Ctrl+Space)', a trial expiration notice 'Trial expires in 14 days', an 'Upgrade' button, a 'Get started faster!' button, and a 'Help' button.

STEP 4: Create Book Details Form

How would you like to create your form?

On my own

+

From scratch

↕

Import with data

Σ

Using Zia

From a template

🚗

Cab Booking

🏠

Conf Room Booking

📅

Appointment Booking

🏨

Hotel Room Booking

🗨

Complaints

50 +

View More

Using an Integrated Datasource

📊

☁

🔗

🔄

📝

...

SIMATS Library Book Reservation ...

Book Details

+

🏠

🔧

⋮

Done

Basic Fields

👤

Name

✉

Email

📍

Address

☎

Phone

📄

Single Line

I

Multi Line

123

Number

📅

Date

📋 Book ID

🗑

📄 Book Title

📄 Author

📄 Publication

📅 123 Year Of Publication

📄 Edition

📄 ISBN Number

📄 Book Category

Field Properties

Field name

Book ID

Field link name

book_id

[View Field References](#)

Validation

Start index

1

Appearance

Field type

Auto Number

🗨

👤

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

📅

📅

Get started faster!

🔗 Help

🗑

STEP 5: Create Members Details Form.

Student ID

Student Name

Registration Number

Department

Year Of Study

Mobile Number

Email

Status

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Field name

Student ID

Field link name

student_id

View Field References

Validation

Start index

1

Appearance

Field type

Auto Number

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

Get started faster! | Help

STEP 6: Create Book Reservation Form.

Student

Book

Reservation Date

Expected Issue Date

Reservation Status

Rack Number

Name

Email

Address

Phone

Single Line

Multi Line

123

Date

Field name

Student

Field link name

student

View Field References

Display Fields

SIMATS Library Book Reservation System → Student Details

Student ID

Display type

Multi Select

Here is your Smart Chat (Ctrl+Space)

Trial expires in 14 days | Upgrade

Get started faster! | Help

STEP 7: Create Reservation Records

Basic Fields

Name	Email
Address	Phone
Single Line	Multi Line
123	15
Number	Date

Field Properties

Field name:

Field link name:

[View Field References](#)

Display Fields: [SIMATS Library Book Reservation System](#) → [Student Details](#)

Student ID + - ÷

Display type:

STEP 10: Add sample data to all the created forms.

Library Book Reservation System

Member Profile

Member Name

First Name

Member ID

Email

Phone Number

Address

Address Line 1

Membership Type

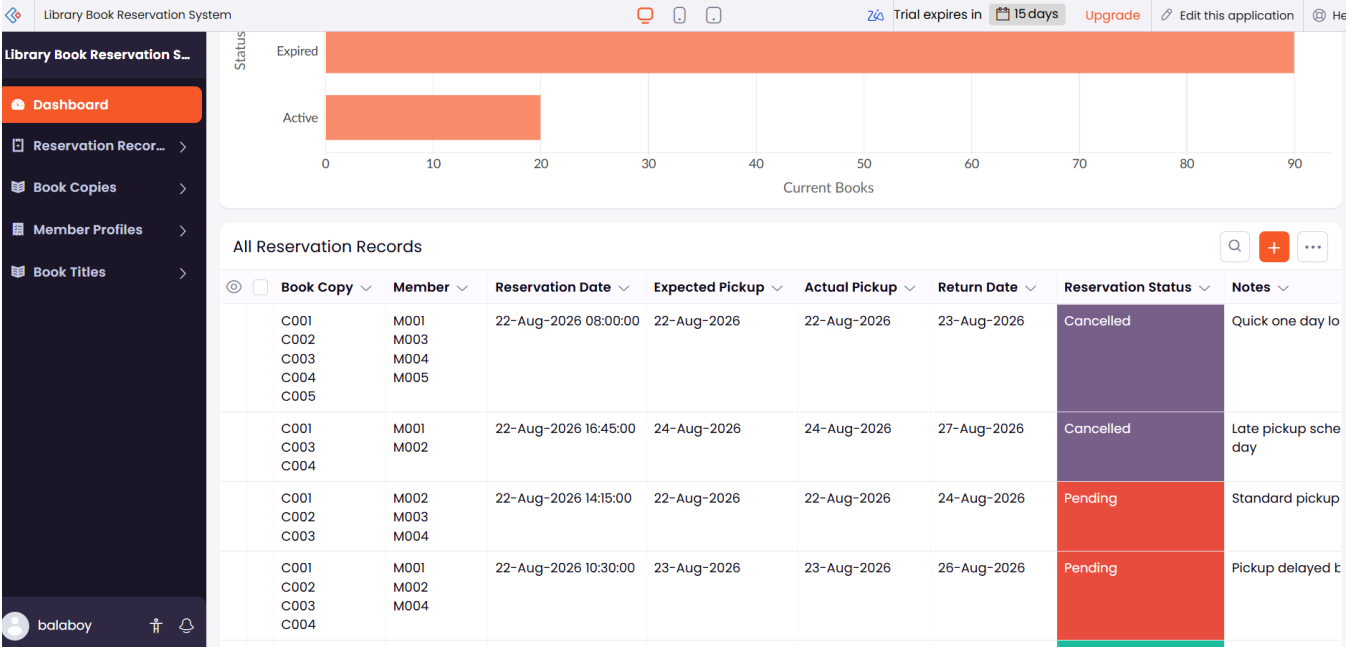
Join Date

Status

Max Books Allowed

Current Books

STEP 11: Create and display the Dashboard



3. Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.

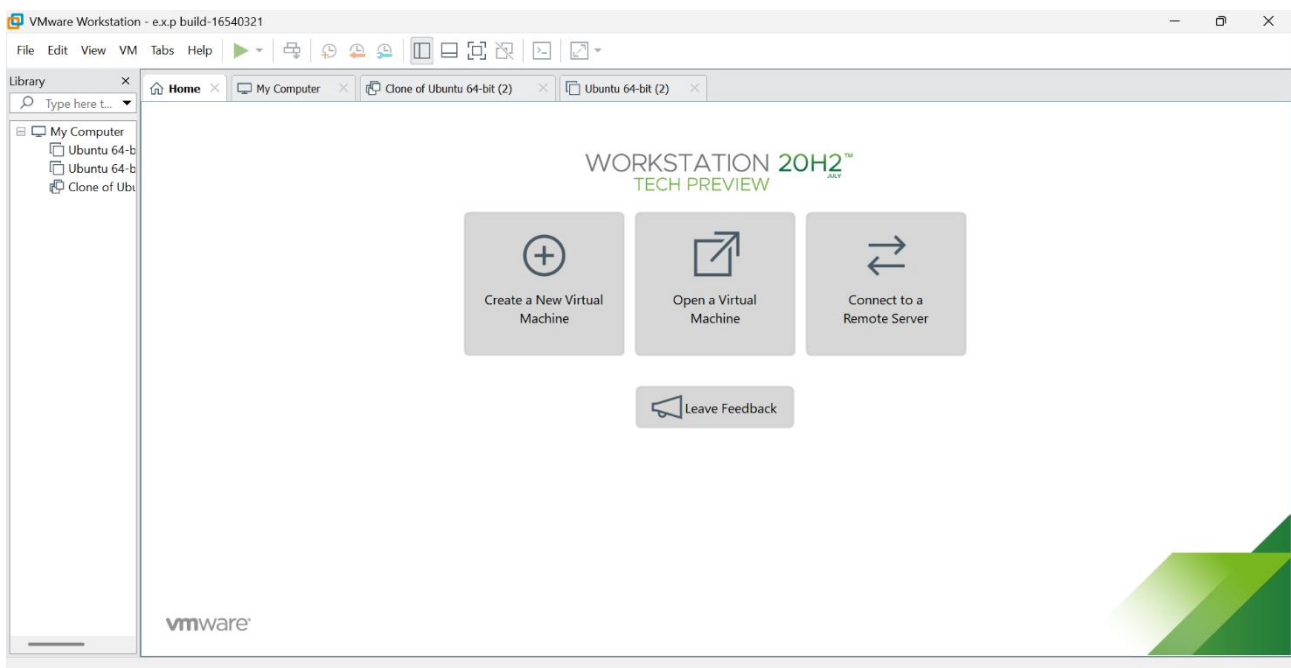
AIM:

Install VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.

Procedure :

1. Install VirtualBox
2. Create the First VM
3. Install Ubuntu
4. Create Another VM
5. Verify the Virtual Machines

Implementation :



New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

C:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-am

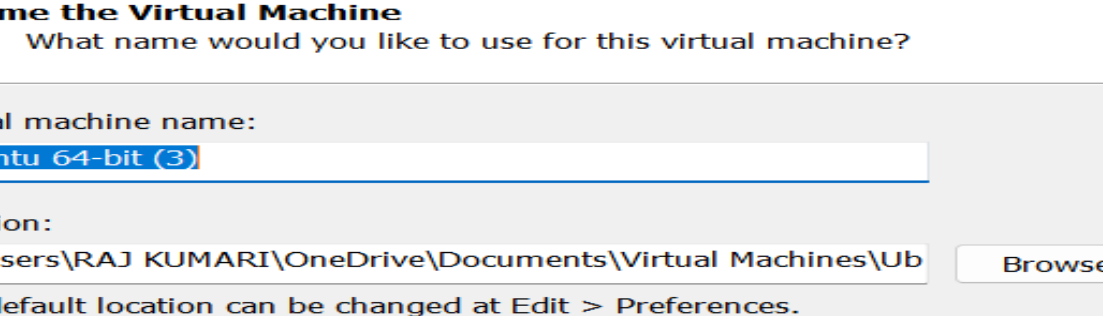
Browse...

 Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. [\(What's this?\)](#)

☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel



New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

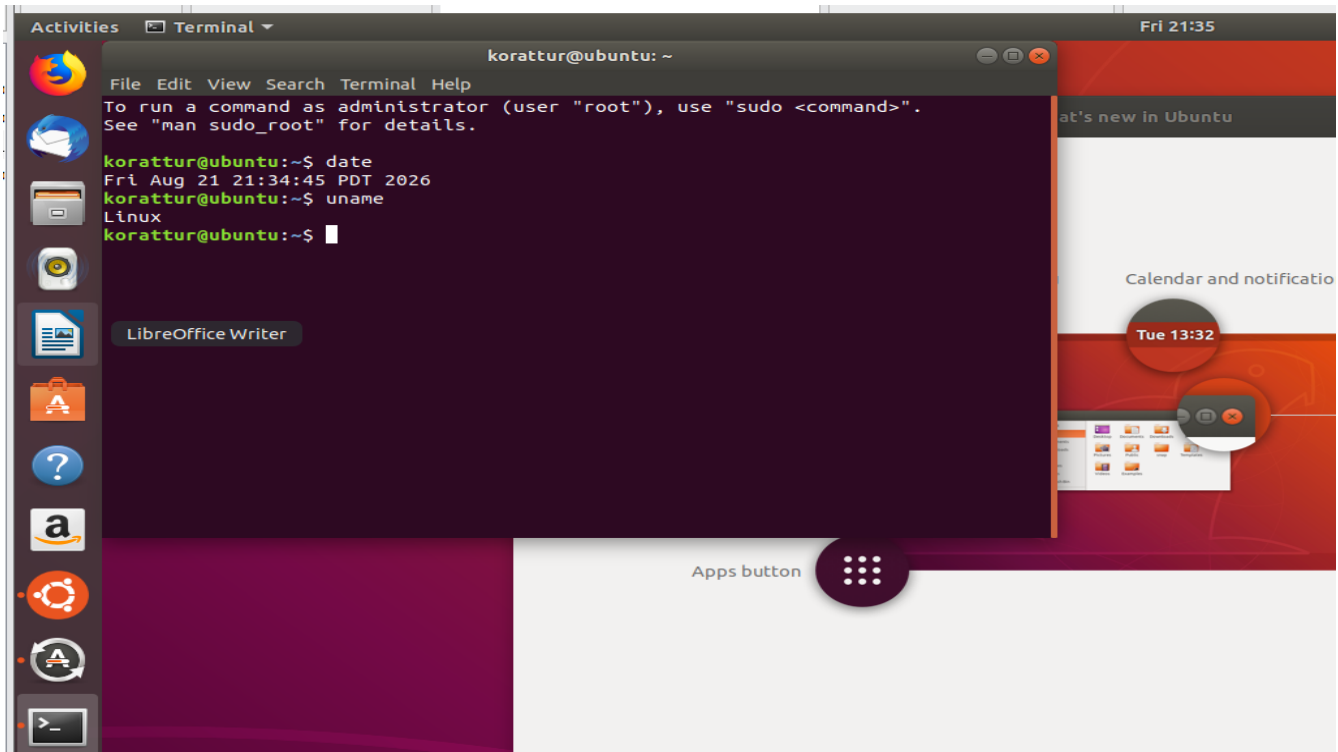
Virtual machine name:
Ubuntu 64-bit (3)

Location:
C:\Users\RAJ KUMARI\OneDrive\Documents\Virtual Machines\Ub

[Browse...](#)

The default location can be changed at Edit > Preferences.

< Back Next > Cancel



Result :

Successfully Installed VirtualBox/VMware Workstation with different flavours of Linux or Windows OS on top of Windows 7 or 8.

4. Showcase virtual machine migration based on certain conditions, from one host to another.

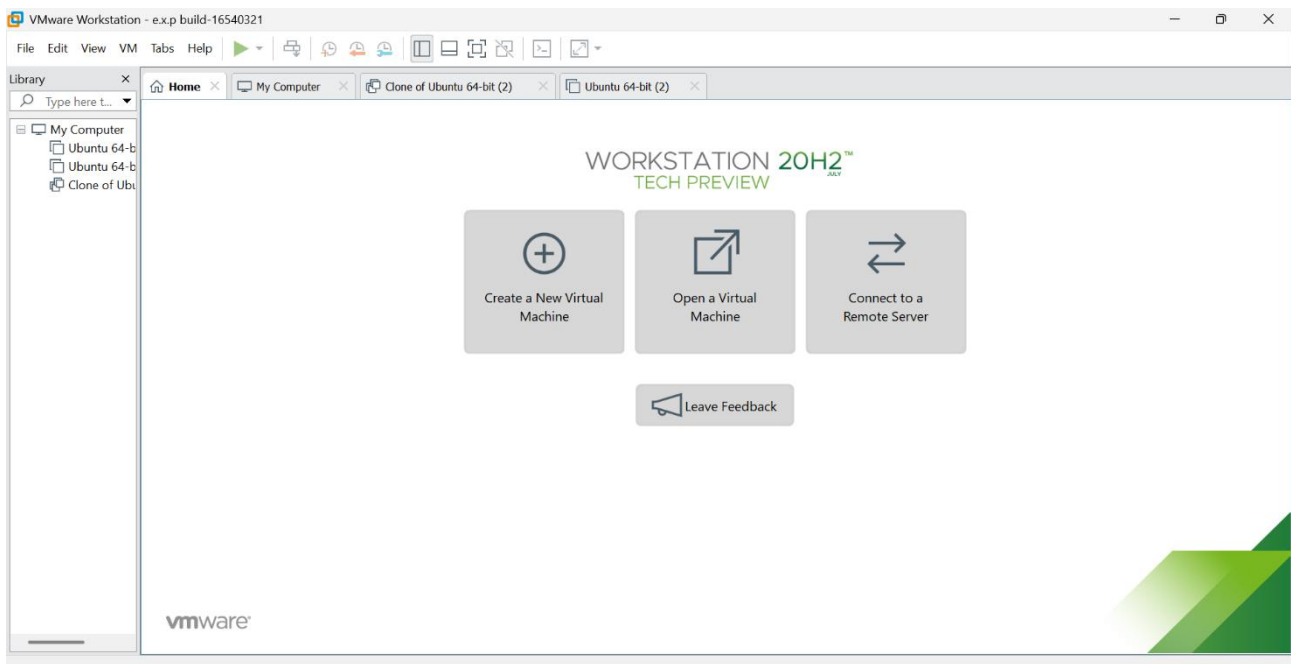
AIM :

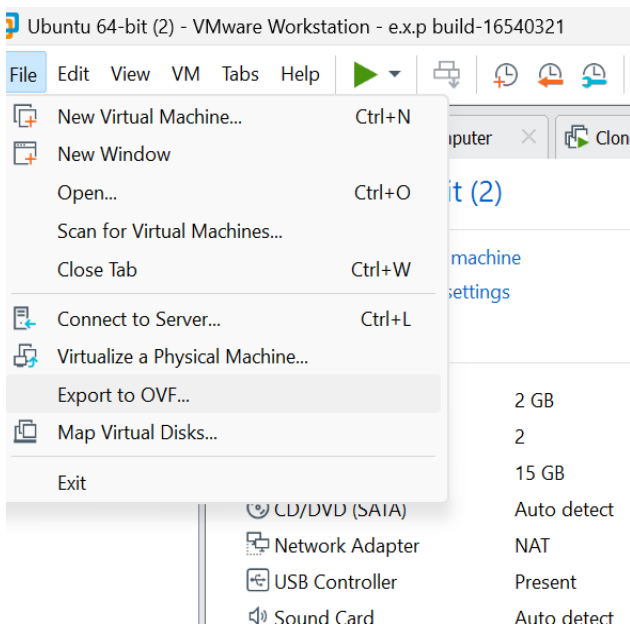
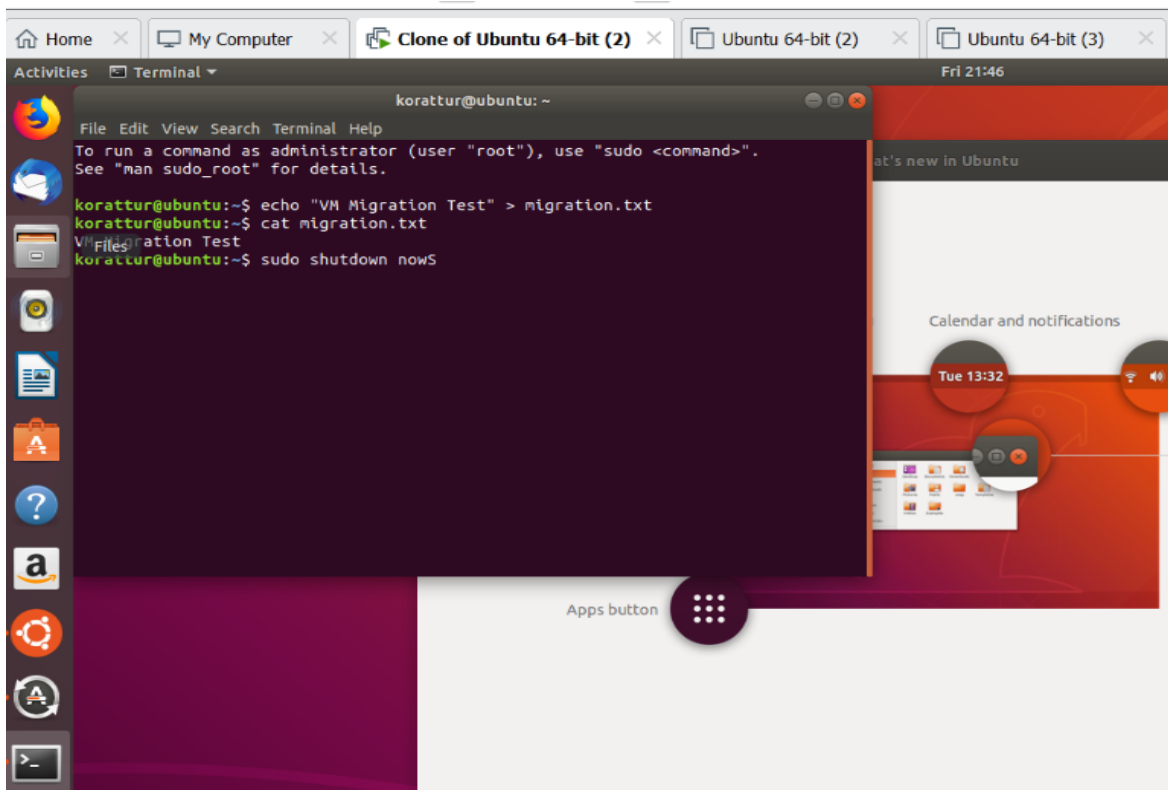
Showcase virtual machine migration based on certain conditions, from one host to another.

Procedure :

1. Create the VM on Host A
2. Create a test file
3. Shut down the VM
4. Export the VM from Host A
5. Import the VM on Host B
6. Start the VM on Host B

Implementation :





Result :

Successfully Showcased virtual machine migration based on certain conditions, from one host to another.

